

Victor Patrin

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SUMMARY

Senior Software Engineer who ships entire products — backend, data pipelines, infrastructure, CI/CD, and AI integration. 6 years at an early-stage AI startup, progressing from intern to leading engineering and reporting directly to the CEO

Building and deploying everything from bare-metal Kubernetes clusters to RAG pipelines to client-facing applications

TECHNICAL SKILLS

Backend: Python, Node.js, FastAPI, Flask, Express.js, SQLAlchemy, Alembic, Pydantic

Data Engineering: PostgreSQL, pgvector, MongoDB, S3, Pub/Sub

DevOps & Cloud: GCP (GKE, IAM), Kubernetes, Helm, Docker, Terraform, Ansible

CI/CD & Observability: GitHub Actions, GitLab CI, Grafana, Prometheus, Loki

Networking & Security: Caddy, HAProxy, Fail2ban, ufw, Trivy, Hadolint, SOPS/age

PROFESSIONAL EXPERIENCE

Senior Software Engineer | GENAIZ, Montreal, Canada

Sep 2022 – Aug 2025

AI-powered automation platform for life sciences, serving enterprise pharma clients

- Led engineering after the CTO's departure — ran sprint planning, product meetings, and daily standups, reported directly to the CEO, and coordinated delivery across a 4-person team
- Saved ~\$20K/year in licensing costs by building a custom Continuous Deployment system (GitLab CI + Helm) to replace Harness CD
- Slashed cloud infrastructure costs by 50% (~\$9K/year) by leading a GCP audit, decommissioning idle assets, and migrating from managed cloud storage to a self-hosted NFS server on Kubernetes
- Owned all GCP infrastructure across 5 Kubernetes clusters (on GKE: dev/staging, prod, 2 dedicated client environments; on a bare-metal cluster for demos), running 15+ microservices — provisioning, scaling, health monitoring, and release coordination
- Deployed the full GENAIZ platform for 3 enterprise pharma clients and built an on-premises bare-metal cluster from scratch for client demos, replacing a dedicated GKE environment
- Part of the quality team leading CFR 21 Part 11 compliance — edited company security policies, drove compliance sprints (audit trails, access controls), and owned policy documentation

Software Engineer | GENAIZ, Montreal, Canada

Oct 2019 – Sep 2022

- Bridged ML research and production — worked with ML engineers to productionalize AI algorithms for pharma clients, including wrapping the NLP pipeline (tokenizer, doc2vec) into Dockerized Python microservices and integrating them into Vespa search
- Reworked ETL pipelines to normalize unstructured pharma documents into a standard format, enabling downstream AI features
- Drove migration of a Node.js monolith into ~8 Python microservices, enabling independent deployment and scaling per service

Blockchain Intern | GENAIZ, Montreal, Canada

Apr – Oct 2019

- Built an immutable audit trail by anchoring event log hashes to Bitcoin transactions for tamper-proof verification

Junior Developer | ubirch GmbH, Berlin, Germany

Jun 2018 – Mar 2019

Blockchain-for-IoT startup

- Developed a mobile application for cryptographic identity verification over Bluetooth BLE and implemented data anchoring on IOTA, Ethereum, and MultiChain networks

PROJECTS

Couquette | AI Wine Discovery Platform | [couquette.club](#) | [App Repo](#) | [Infra Repo](#)

Feb 2026 – Present

- AI sommelier for Quebec's SAQ catalog — bilingual (FR/EN) wine recommendations via a React frontend, a Telegram bot, and a FastAPI backend (PostgreSQL + pgvector)
- Hybrid RAG pipeline: Claude intent routing → SQL filters → pgvector cosine similarity → MMR re-ranking → Claude Haiku curation
- LLM eval framework: multi-query test suite (FR/EN, train/holdout), 5-dimension rubric (relevance, diversity, coherence, curation, adherence), Claude Sonnet as judge — tracked improvement from 3.37 → 4.05/5 across iterative optimization cycles
- Multi-source scraper (SAQ sitemap, BeautifulSoup, Adobe Live Search) with robots.txt compliance and rate limiting
- Full production platform in a separate infrastructure repo — IaC end-to-end: Terraform (DNS, firewall), Ansible (server configuration, CIS hardening), disaster recovery runbook for full redeployment from scratch, Grafana/Loki/Prometheus observability, automated PostgreSQL backups to S3 (30-day retention)
- CIS/ANSSI hardening (Lynis 80/100, testssl.sh A+), container hardening (cap_drop, read-only fs, no-new-privileges), secret management (sops/age)

EDUCATION

M.Sc. Information & Systems Security | Université de Lorraine, France

2018 – 2019

Diplôme d'Ingénieur (M.Eng.) | Mines de Nancy, France

2016 – 2019

Classe Préparatoire MPSI/MP | Lycée Gay-Lussac, France

2013 – 2016

INTERESTS

Ultra-endurance cycling · Wine tasting ([WSET Level 3](#) candidate) · Philosophy